
AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong & Ozdaglar (2026)

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Deep-dive and oral-exam backup

github.com/carlosgomezp28/ai-04-acemoglu

Research question and mechanism

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How does agentic AI change human learning incentives and the long-run stock of collective knowledge?

HUMAN EFFORT

private context signal
+
thin public signal



TWO RETURNS

private task success
+
future public knowledge



AGENTIC AI

context-specific precision
substitutes for the
private motive to learn

The externality is the hinge: short-lived agents receive the private return from learning, but do not internalize the knowledge inherited by later cohorts.

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Latent states

$$\begin{aligned}\theta_{t+1} &= \theta_t + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, \Sigma^2), \\ \theta_{i,t} &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).\end{aligned}$$

Signals

$$\begin{aligned}s_{i,t}^H &\sim \mathcal{N}(\theta_{i,t}, (\lambda_I e_{i,t})^{-1}), \\ s_{i,t}^A &\sim \mathcal{N}(\theta_{i,t}, \tau_A^{-1}), \\ s_t^C &\sim \mathcal{N}(\theta_t, (\lambda_G l e_t)^{-1}).\end{aligned}$$

Posterior precisions

$$X_t = \text{Var}(\theta_t \mid \mathcal{I}_t)^{-1},$$

$$Y_{i,t} = \sigma^{-2} + \lambda_I e_{i,t} + \tau_A.$$

X_t is inherited, public, general knowledge.

$Y_{i,t}$ combines prior precision, human learning, and the agentic-AI recommendation for the current context.

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Four task outcomes, re-parameterized

$$\Delta_G = f(1, 0) - f(0, 0),$$

$$\Delta_I = f(0, 1) - f(0, 0),$$

$$\Delta_X = f(1, 1) - f(1, 0) - f(0, 1) + f(0, 0).$$

The normalization $f(1, 1) - f(0, 0) = 1$ gives

$$\Delta_G + \Delta_I + \Delta_X = 1.$$

Assumption 1

$$\Delta_I = 0, \quad \Delta_X > 0.$$

$\Delta_I = 0$ is strong: correct context-specific knowledge has no incremental productive value unless general knowledge is also correct.

$\Delta_X > 0$ supplies strict complementarity; Δ_G may be positive.

Static expected utility

PAPER RESULT Let $p_G = G(X)$ and $p_I = G(Y)$ be the two Gaussian success probabilities. The four production outcomes imply the accounting identity

$$\mathbb{E}[f] = f_{00} + \Delta_G p_G + \Delta_I p_I + \Delta_X p_G p_I.$$

Under the paper's maintained $\Delta_I = 0$,

$$U(e; X, \tau_A) = f_{00} + \Delta_G G(X) + \Delta_X G(X)G(Y) - \frac{e^\alpha}{\alpha}$$

$$Y = \sigma^{-2} + \lambda_I e + \tau_A, \quad \alpha > 1.$$

The individual treats X as given: her infinitesimal public-learning contribution benefits future cohorts and is not privately internalized.

First-order condition and uniqueness

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$$U_e = \underbrace{\Delta_X G(X) \lambda_I g(Y)}_{\text{marginal information benefit}} - \underbrace{e^{\alpha-1}}_{\text{marginal effort cost}} .$$

$$\Delta_X G(X) \lambda_I g(\sigma^{-2} + \lambda_I e + \tau_A) = e^{\alpha-1}$$

Since g is decreasing, marginal benefit weakly falls with e . Since $\alpha > 1$, marginal cost strictly rises. Their crossing is unique and finite.

At $X = 0$, $G(0) = 0$: the information benefit vanishes and the paper's unique best response is $e(0, \tau_A) = 0$.

Observation 1: general knowledge

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$$U_e = \Delta_X G(X) \lambda_I g(Y) - e^{\alpha-1}.$$

Differentiate the marginal return to effort with respect to public precision:

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0.$$

Conditions for the strict sign:

$$\Delta_X > 0, \quad \lambda_I > 0, \quad X > 0, \quad Y > 0, \quad g(X) > 0, \quad g(Y) > 0.$$

More general knowledge makes context-specific evidence more useful; that raises the marginal return to acquiring one's own signal.

Observation 1: agentic AI

PAPER RESULT Since $\partial Y / \partial \tau_A = 1$,

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$

Conditions for the strict sign:

$$\Delta_X > 0, \quad \lambda_I > 0, \quad X > 0, \quad Y > 0, \quad G(X) > 0, \quad g'(Y) < 0.$$

more AI precision τ_A $\implies$ lower marginal gain from self-acquired precision $\implies$ lower human effort

The welfare trap

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STATIC — hold X fixed

$$\tau_A \uparrow \Rightarrow G(Y) \uparrow$$

More accurate advice raises the private value of the current decision.

DYNAMIC — equilibrium feedback

$$\tau_A \uparrow \Rightarrow e_t \downarrow \Rightarrow X_{t+1} \downarrow .$$

Less human learning weakens the public knowledge inherited by later cohorts.

Under Assumption 2, $\sigma^{-2} \geq \sqrt{2} - 1$:

- ▶ if $\alpha - 1 > 1/4$, Proposition 10 gives a finite welfare peak τ_A^* (possibly zero);
- ▶ if $\alpha - 1 < 1/4$, Proposition 11 gives the analogous peak below the collapse threshold τ_A^c , and sufficiently accurate AI can eliminate the high-knowledge steady state.

A positive static AI effect does not imply monotone long-run welfare.

What Section 5 relaxes

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Section	Baseline object changed	Main qualitative consequence
5.1 AI-enhanced aggregation	Fixed I becomes $I(\tau_A) = I_0 + \exp(\eta\tau_A)$.	Substitution versus aggregation. If $\eta < 1/[2(\alpha - 1)]$, the high state still tends to zero as $\tau_A \rightarrow \infty$.
5.2 Synthetic data	Add exogenous public precision $\tau_{\text{syn}} \geq 0$.	$\tau_{\text{syn}} > 0 \Rightarrow F_{\text{syn}}(0) > 0$: the low state is strictly positive.
5.3 Imperfect separability	Public precision scales with e^β rather than e .	For $\beta > 0$, baseline logic persists; the stability cutoff becomes $\alpha - 1 < \beta/4$.

All three extensions change information creation or aggregation.

What the paper never relaxes

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$$\Delta_I = 0, \quad \Delta_X > 0$$

Section 5 changes

- ▶ aggregation capacity I ;
- ▶ exogenous public precision τ_{syn} ;
- ▶ mapping from effort into public signals, e^β .

Section 5 does not change

- ▶ the task-production function f ;
- ▶ zero standalone value of context-specific success;
- ▶ strict production complementarity.

This is the production-side trap.

Proposed extension: standalone context-specific value

OUR PROPOSED EXTENSION — NOT IN THE PAPER

Four-outcome expected payoff

$$U = f_{00} + \Delta_G G(X) + \Delta_I G(Y) + \Delta_X G(X)G(Y) - \frac{e^\alpha}{\alpha}.$$

The new term $\Delta_I G(Y)$ makes correct context-specific knowledge productive even if general knowledge is wrong.

Assumptions

$$\Delta_I > 0, \quad \Delta_X > 0,$$

$$\Delta_G = 1 - \Delta_I - \Delta_X \geq 0.$$

Define

$$B(X) = \Delta_I + \Delta_X G(X) > 0.$$

Then the modified FOC is

$$B(X)\lambda_I g(Y) = e^{\alpha-1}.$$

Baseline versus extension at $X = 0$

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$$\Delta_I = 0 \text{ and } G(0) = 0$$

$$\Rightarrow U_e = -e^{\alpha-1} \Rightarrow e^*(0, \tau_A) = 0.$$

Hence the paper's transition satisfies
 $F(0) = 0$.

OUR PROPOSED EXTENSION — NOT IN THE PAPER

$\Delta_I > 0, \lambda_I > 0, \alpha > 1$, finite precisions

$$\Rightarrow e^*(0, \tau_A) > 0.$$

If also $\lambda_G > 0, I > 0$, and $0 \leq \Sigma^2 < \infty$,

$$F_{\Delta_I}(0) = [(\lambda_G I e_0^*)^{-1} + \Sigma^2]^{-1} > 0.$$

Exact zero public knowledge is no longer a fixed point for finite AI precision. Severe low-knowledge outcomes may remain.

Numerical illustration and compiled Lean checks

OUR PROPOSED EXTENSION — NOT IN THE PAPER

Audited calibration

$$\alpha = 2, \quad \lambda_I = \lambda_G = I = \Sigma^2 = 1,$$

$$\sigma^{-2} = 0.5, \quad \Delta_I = 0.25,$$

$$\Delta_X = 0.5, \quad \Delta_G = 0.25.$$

X	τ_A	e^*
0	0	0.09591
0.25	0	0.15653
0	2	0.01785

$$F(0) \approx 0.08752 > 0.$$

Compiled Mathlib verification

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- ▶ general-knowledge / effort sign;
- ▶ AI / effort sign.

OUR EXTENSION

- ▶ positive $B(X)$ and positive boundary marginal benefit;
- ▶ $F_{\Delta_I}(0) > 0$ under added positivity assumptions.

Exact theorem names: `lean/Main.lean`

Lean does NOT formalize Gaussian conjugacy/derivatives, the optimization problem, equilibrium, or the full dynamic model.

Where I did not believe AI: final audit

6. Implication for public knowledge

The baseline transition law is

$$X_{t+1} = [(X_t + \lambda_0 I e^* + \varepsilon^2)^+]^+$$

At $X=0$

$$F_{\Delta_I}(0) = [(\lambda_0 I e^*(0, \tau_A))^+ + \varepsilon^2]^+ > 0$$

provided

$$\lambda_0 > 0, \quad I > 0$$

7. Final result

$\Delta_I > 0 \rightarrow F_{\Delta_I}(0) > 0$

Result. Positive standalone value of context-specific knowledge removes the exact zero-knowledge fixed point for finite AI precision

It doesn't imply that knowledge collapse disappears in general

Handwritten audit, p. 3

OUR PROPOSED EXTENSION — NOT IN THE PAPER

INITIAL CLAIM

$\Delta_I > 0 \Rightarrow$ knowledge collapse is eliminated.

AUDIT: TOO STRONG.

CORRECTED CONCLUSION

$\Delta_I > 0 \Rightarrow F_{\Delta_I}(0) > 0$

under the relevant static and public-learning conditions.

Exact zero disappears, but low positive states, multiplicity, path dependence, and welfare loss may remain.

Synthetic data: exogenous public precision; our mechanism: endogenous human effort.

OUR INTERPRETATION The paper's collapse mechanism is robust to many information-side extensions, but its exact zero state depends critically on assuming that context-specific knowledge has no standalone productive value.